Optimizing Radish Crop Yield Using a Combinatorial Taguchi–Grey Relational Analysis Approach

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Abstract

The globally cultivated root vegetable, radish, has a notably quick growth cycle of three to four weeks. This rapid maturation makes it highly suitable for early planting and for use in intercropping systems. There is a recognized deficit in research regarding optimal agricultural practices in growing radish. This is largely because radish is often categorized as a minor crop. Agricultural productivity continues to be challenged by issues such as low yields and the inefficient use of resources. Traditional approaches to determining the best combination of agricultural inputs are often both time-consuming and resource-intensive. To address these limitations, a statistical method known as Design of Experiments, particularly when utilizing Taguchi's Orthogonal Arrays, provides a powerful framework for systematically optimizing multiple factors that influence crop yield. This method efficiently identifies the ideal input levels needed to maximize yield, while simultaneously reducing the number of experimental trials and resource consumption, thereby fostering more sustainable agricultural practices. This study used radish as the experimental crop to investigate how three key variables—seed variety, irrigation method and manure composition - affect yield. Using Taguchi's Orthogonal Arrays, nine unique treatment combinations were developed, which were then applied to randomized experimental plots, each with two replications. Following the experimental period, crop yield was measured by quantity, weight, and length. Grey Relational Analysis was subsequently employed to pinpoint the most effective combination of variables for achieving maximum yield of radish. Confirmation experiments were carried out using the optimum levels identified for the variables and significant improvement in the yield was validated.

Keywords: *Radish yield, Seed, Manure, Irrigation, Design of Experiments, Orthogonal Array, Grey Relational Analysis*

Introduction:

Radish is a rapidly maturing root crop with a short growth cycle, making it well-suited for early-season cultivation and intercropping systems. In addition to its nutritional benefits, it enhances soil structure through aeration and supports sustainable farming when used in crop rotation. Cultivated widely across Asia, Europe, and the Americas, the radish stands out for its adaptability and contribution to global food security and dietary variety. Modern agriculture, however, continues to face challenges, including limited yields, inefficient resource use, and negative environmental impacts. Conventional farming methods often depend on intensive labor and time-consuming trial-and-error approaches to determine the right nutrient combinations for boosting productivity. This process frequently produces less-than-optimal results. To overcome these issues, this paper explores the applicability of Design of Experiments (DOE)—a systematic statistical methodology for designing and analysing experiments efficiently, and Grey Relational Analysis (GRA) for analysing the data obtained from the experiments. By varying multiple input factors in a structured way and studying their

influence on outcomes, DOE reveals key drivers of radish yield, resource efficiency, and ecological impact. These insights provide a foundation for more informed decisions by farmers and other stakeholders. This study reflects a transition from traditional trial-based methods to a data-driven framework for agricultural improvement. Through the use of DOE, the authors seek to deliver practical recommendations that support better crop management and sustainability.

Literature Review:

Radish is a widely cultivated root vegetable, valued for its versatility and global distribution. Despite its relevance in agriculture, research focusing on the determinants of radish yield has been relatively limited, likely due to its classification as a minor crop. The productivity of radish is shaped by several agronomic factors such as seed quality, fertilizer use, weed control, farmer education, and plant protection practices. Bakhsh et al. (2006) observed that although radish can adapt to a range of soil conditions, it performs best in light, fertile, and well-moist soils. Planting density has also been reported to significantly influence output, with higher densities improving yield per acre. Proper application of nitrogen and phosphorus fertilizers further enhances productivity, while effective weed management is essential to prevent substantial losses from competition with weeds. These findings reinforce the need for integrated crop management practices to achieve optimal yields. Earlier studies, including those by Ahmad et al. (2003) and Bakhsh et al. (2004), highlight the major roles of seed quality and fertilizer in determining vegetable crop performance. However, Lúcio and Sari (2017) point out that experiments involving vegetables are often complex due to intensive management demands and high labor inputs, and therefore emphasize strategies for reducing experimental variability. More recent work by Shilpa et al. (2022) has evaluated combinations of organic, inorganic, and bio-fertilizers, showing their combined influence on radish growth, yield, quality, and soil health. Such comprehensive studies contribute to better understanding of nutrient management strategies. Similarly, research addressing resource use efficiency stresses its importance for sustainable productivity. For instance, Shrabanti et al. (2023) report that inefficiencies in fertilizer and pesticide use negatively affect farm outputs, while Effiong (2005) notes that the high cost of inputs particularly burdens smallholders, reducing their ability to utilize resources efficiently. Improving resource management, especially in staple crop production, has the potential to reduce hunger, alleviate unemployment, and support broader societal well-being. In this context, the reviewed literature has been collated and summarized below.

Effect of Nitrogen, Phosphorus, Potassium on Radish Yield

The productivity of radish is strongly dependent on the availability of key macronutrients, namely nitrogen (N), phosphorus (P), and potassium (K). A study by Pujankhanal et al. (2022) examined the influence of different levels of these nutrients on the variety *Pusa Himani*, identifying 80:40:60 kg/ha of N, P, and K, respectively, as the most effective combination for achieving higher yields. Nitrogen plays a fundamental role in plant growth, being directly involved in protein and protoplasm formation, which supports cell division and overall vegetative development (Lakra et al., 2017). Deficiency of this nutrient often results in reduced quality and lower yields at harvest (Hussain et al., 1977). Phosphorus is equally important, as it is integral to several physiological processes and is particularly critical for root growth. Insufficient phosphorus leads to symptoms such as stunted plants and reddish-purple

discoloration of leaves, ultimately reducing yield potential (Zeb et al., 2016). Potassium, the third essential macronutrient, supports photosynthesis, food translocation, and protein synthesis, thereby contributing significantly to healthy growth and crop productivity (Inam et al., 2011). Ensuring balanced and sufficient levels of nitrogen, phosphorus, and potassium in the soil is therefore vital for achieving optimum radish growth, yield, and quality.

Irrigation Methods

Proper irrigation management is crucial for achieving high radish yields while ensuring efficient use of water. Rai et al. (2017) compared different irrigation methods, such as surface, sprinkler, and drip irrigation, noting the advantages and drawbacks of each. Among these, drip irrigation has been recognized as particularly effective because it supplies water directly to the root zone, thereby reducing losses through evaporation and runoff (Saxena, 2021). This method not only conserves water but also improves nutrient absorption by enabling fertilizers to be applied efficiently at the root level. The adoption of modern systems such as sprinkler and drip irrigation has increased in recent years, especially among farmers aiming to cope with water shortages and improve productivity (Jahangirpour & Zibaei, 2022). These technologies allow for precise regulation of water delivery according to crop requirements and environmental conditions. By integrating such advanced practices, farmers can enhance water-use efficiency, minimize wastage, and sustain higher yields even under conditions of limited water availability.

DOE in Agriculture

DOE has emerged as an effective methodological framework for optimizing agricultural practices and improving productivity. As noted by Durakovic (2017), DOE is highly versatile and applicable across diverse fields, including agriculture, where it supports tasks such as variable screening, process optimization, and robust system design. In the context of crop production, precision agriculture tools have enabled the use of on-farm field trials to generate reliable and cost-effective data, particularly for refining fertilizer management strategies (Bullock et al., 2020). Such field-based experimentation allows farmers to directly assess the performance of different fertilization regimes and adjust nutrient inputs to align with specific soil conditions and crop requirements. Sahu et al. (2018) applied a Randomized Block Design to evaluate integrated nutrient management practices, demonstrating that combinations of different organic manures significantly influenced radish growth across developmental stages. Similarly, Taguchi-based designs have been adopted in bioprocess optimization, with Marchut-Mikołajczyk et al. (2021) and Inès Mnif et al. (2021) presenting innovative applications of this approach. Beyond nutrient management, DoE also provides a structured means of determining optimal planting densities, irrigation regimes, and broader crop management strategies through systematic experimentation and statistical analysis (Bullock et al., 2020). Overall, the application of DOE in agricultural research and practice represents a shift toward data-driven decision-making. By identifying key factors and their interactions, this methodology enables farmers and researchers to enhance yields, improve resource-use efficiency, and reduce environmental impacts, ultimately contributing to more sustainable agricultural systems.

Taguchi's Orthogonal Array Approach in Agriculture

The Taguchi method offers a structured framework for process optimization and quality enhancement within agricultural systems. Fraley et al. (2023) emphasized the value of Taguchi orthogonal arrays in minimizing process variation and identifying critical factors that influence product quality. In contrast to conventional factorial designs, the Taguchi approach employs orthogonal arrays to efficiently evaluate parameter interactions and determine optimal operating conditions, thereby reducing the number of experimental trials required while still generating robust insights into system behaviour. This methodology has been increasingly applied in agriculture to optimize diverse processes, including pyrolysis and mixture formulations (Canales Flores et al., 2017; Singh et al., 2020). Through systematic variation of input parameters and careful analysis of experimental outcomes, Taguchi designs facilitate the identification of conditions that enhance product quality and improve resource-use efficiency. The methodological efficiency and practical applicability of the Taguchi method thus make it a valuable tool for advancing process optimization and sustainability in agricultural production.

GRA in Agriculture

GRA provides a valuable framework for decision-making in agricultural management, particularly under conditions of uncertainty and incomplete information. Xie et al. (2009) demonstrated its effectiveness in assessing complex systems by quantifying relationships among multiple factors, thereby supporting more informed decisions and efficient resource allocation. Applications of GRA in agriculture include crop selection, land-use planning, and broader resource management. The integration of Taguchi methods with GRA has been shown to further strengthen process optimization and quality improvement efforts in agriculture. According to Thapa and Engelken (2020), the combination of these two techniques allows researchers to identify optimal process parameters while simultaneously addressing system variability. This combined approach not only enhances product quality but also improves efficiency in resource utilization. In the context of radish cultivation, achieving higher yields depends on effective nutrient management, appropriate irrigation strategies, and the application of advanced experimental techniques. The use of Taguchi orthogonal arrays in conjunction with GRA offers considerable potential for improving productivity, optimizing resource use, and supporting evidence-based decision-making. Overall, the reviewed literature underscores the importance of resource optimization as a central factor in achieving sustainable efficiency and maximizing agricultural output.

The objectives of this paper are to identify the most influential factors affecting radish growth and yield, and determine the optimum combination of key factors that promote maximum radish growth and yield.

Experimentation and Data Collection:
Radishes were chosen for the DOE because of their short growth cycle, which allows rapid data collection, and their ease of cultivation, which facilitates controlled experimental conditions. Their genetic diversity makes them suitable for studying a range of agronomic traits relevant to crop improvement, while their role in crop rotation offers valuable insights into sustainable farming practices. Owing to their versatility and rapid growth, radishes serve as an effective model crop for exploring soil health, crop management, and genetic diversity in agricultural research. Radish yield is determined by a combination of environmental and management factors. Key influences include soil characteristics (such as composition, pH, and fertility), climatic conditions (temperature, rainfall, and sunlight), and cultural practices like planting density, nutrient application, and irrigation management. Effective pest, disease, and

weed control, along with appropriate pollination and harvesting techniques, are also essential for achieving maximum yield. For the present study, three major factors were selected for investigation: seed variety (A), irrigation method (B), and manure composition (C). These were chosen due to their well-documented impact on radish production. Seed variety governs the genetic potential for growth, stress tolerance, and adaptability. Irrigation method determines water availability and nutrient uptake, both of which are critical for root crop development. Manure composition contributes essential nutrients and enhances soil structure, directly influencing plant vigour and yield. The levels chosen for each of these factors are outlined below.

Seed variety - 3 Levels [3 types of brands were selected]
Brand I: Sungrow seeds
Brand II: Trusted seeds
Brand III: Keyonic seeds

Irrigation method - 2 Levels [2 types of irrigation]
Level 1: Drip irrigation
Level 2: Sprinkler irrigation

Manure composition - 3 Levels [3 types of manure]
Brand I: DAP manure
Brand II: 20-20-20
Brand III: Organic manure

Accordingly, L9 orthogonal array has been selected for experimentation and the experimental layout is shown in table 1.

Table 1: L9 Orthogonal array layout and experimental layout

Expt. No.	L9 orthogonal array layout				Experimental layout			
	1	2	3	4	A-Seed variety	B-Irrigation method	C-Manure composition	D*
1	1	1	1	1	Sungro	Drip	DAP	
2	1	2	2	2	Sungro	Sprinkler	20-20-20	
3	1	3	3	3	Sungro	Drip*	Organic	
4	2	1	2	3	Trusted Seeds	Drip	20-20-20	
5	2	2	3	1	Trusted Seeds	Sprinkler	Organic	
6	2	3	1	2	Trusted Seeds	Drip*	DAP	
7	3	1	3	2	Keyonic Seeds	Drip	Organic	
8	3	2	1	3	Keyonic Seeds	Sprinkler	DAP	
9	3	3	2	1	Keyonic Seeds	Drip*	20-20-20	

* Drip is selected for level 3 of factor B

This experimentation was carried out in Mandya District, KR Pete Taluk , Kikkeri Hobli, Echalaguday, Karnataka, with the Google maps location link: https://maps.app.goo.gl/YXKmzfPKwx8ghEm79?g_st=iw.

Prior to initiating the experiment, the land designated for radish cultivation was carefully evaluated for its suitability. The soil consisted of a heterogeneous mix of red loamy, sandy, and black soils, offering varied physical and chemical properties. The site received adequate sunlight, with 6–8 hours of full sun exposure daily, and the average ambient temperature was approximately 30°C. Water availability was sufficient, supported by both a bore well and the nearby Hemavathi River, ensuring irrigation for a minimum of five hours per day. The presence of mature coconut, areca palm, and jackfruit trees on the site was also noted, as these could influence the immediate microenvironment. Before planting, the field was systematically prepared. Debris, rocks, and existing vegetation were removed, and the soil was tilled to a depth of 6–8 inches using a tractor. Raised beds were then formed and levelled with garden rakes, dividing the area into sections suitable for sowing. The experimental design was established by marking 18 square plots, An irrigation system was subsequently installed: six lateral pipes with integrated drippers supplied 12 of the plots, while one lateral pipe equipped with a jet sprinkler (shown in figure 1) irrigated the remaining six plots.



Figure 1: Drip and sprinkler irrigation systems

Radish seeds and manure were applied using the broadcast method to ensure uniform distribution across the plots. Weed control was carried out manually by hand-pulling to eliminate unwanted grasses and competing vegetation. Harvesting was conducted plot-wise, with yields from each plot collected separately in labelled crates to avoid cross-mixing. After harvest, the radishes were washed thoroughly, excess plant material was trimmed, and the roots were air-dried for one hour before leaf separation. For each of the 18 experimental plots, yield parameters were systematically recorded. Measurements included radish length (cm) using a measuring tape, the total number of radishes harvested, and the total fresh weight (kg) determined with a digital weighing scale. The experimental results are summarized in Table 2, while the images of the harvested radishes are provided in Figure 2.

Table 2: Experimental Results

Expt. No.	Experimental layout				Responses		
	A-Seed variety	B-Irrigation method	C-Manure composition	D*	Average Length (cm)	Average Quantity	Average Weight (kg)
1	Sungro	Drip	DAP		10.41	37.5	1.2
2	Sungro	Sprinkler	20-20-20		12.58	38.5	1.6

3	Sungro	Drip*	Organic		9.74	36.0	0.7
4	Trusted Seeds	Drip	20-20-20		12.07	41.0	1.9
5	Trusted Seeds	Sprinkler	Organic		10.90	43.0	1.0
6	Trusted Seeds	Drip*	DAP		14.75	35.5	2.3
7	Keyonic Seeds	Drip	Organic		12.74	45.0	1.2
8	Keyonic Seeds	Sprinkler	DAP		13.71	28.0	1.4
9	Keyonic Seeds	Drip*	20-20-20		13.84	24.5	1.0

*empty column



Figure 2: Images of the harvested radishes for each treatment

Analysis of Experimental Results

S/N ratio analysis was carried out using Minitab statistical software for the weight, quantity and length of the radishes using Taguchi S/N ratio for larger-the-better, given in Eq. (1), since the

objective is to maximize the weight, length and quantity.

S/N ratio (larger-the-better) = $-10 * \log_{10}(\Sigma(1/y^2) / n)$ ….Eq.(1)

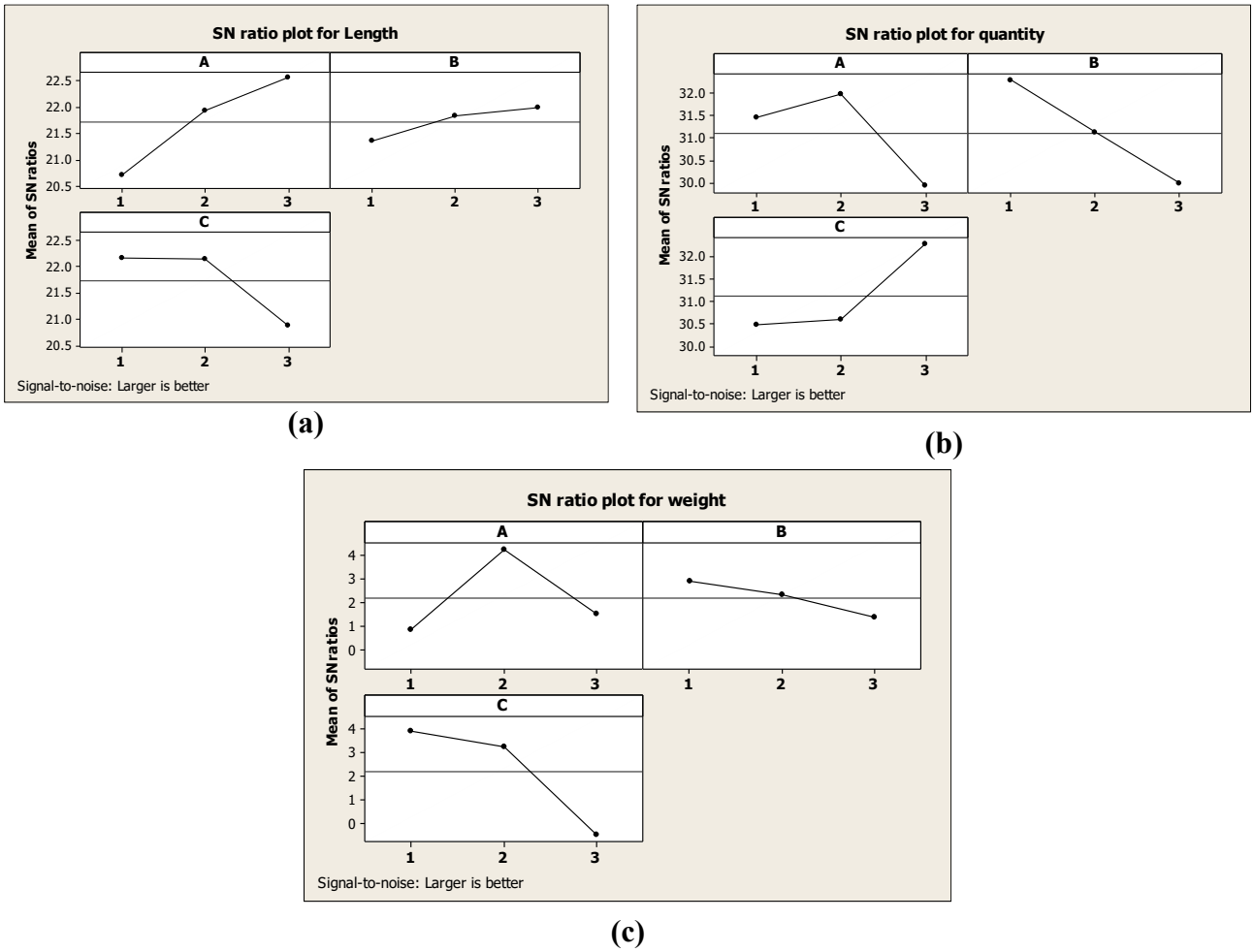


Figure 3: (a) S/N ratio plot for length (b) S/N ratio plot for quantity (c) S/N ratio plot for weight

The S/N ratio plots for length, quantity and weight of the radish are shown in figure 3. For the radish length, the optimal combination is Keyonic seeds (A), Drip irrigation (B) and DAP manure (C). For the quantity of radishes, the optimal treatment combination is trusted seeds (A), drip irrigation (B) and organic manure (C). For the weight of the radish, the optimal combination is Trusted seeds (A), Drip irrigation (B) and DAP manure (C). Since the optimal factor level combination varies for each response, GRA has been employed to determine the overall optimum levels across all the responses.

The experimental results shown in table 2 are normalized using the equation (2).

$$x_i(k) = \frac{y_i(k) - \min y_i(k)}{max\ y_i(k) - \min y_i(k)} \quad \text{….Eq.(2)}$$

for each experimental trial are shown in table 3.

$\Delta_{0i}(k) = 1 - x_i(k)$ ……Eq (3)

$\zeta\,(\mathrm{k}) = \frac{\Delta\min + £\Delta max}{\Delta_{oi}(\mathrm{k})\min + £\Delta max}$ -----Eq (4)

The deviation sequences for the normalized values and the Grey Relational Coefficients are calculated using equations 3 and 4 respectively. The Grey Relational Grades (GRG) are obtained by averaging the corresponding coefficients for each treatment. The normalized values, deviation sequences, coefficients, GRG along with the ranks for each experiment are shown in table 3.

Table 3: Normalized values, Deviation sequences and Grey Relational Coefficients

Expt. No.	Normalized Values			Deviation Sequences			Coefficients			GRG	Rank
	Length	Quantity	Weight	Length	Quantity	Weight	Length	Quantity	Weight		
1	0.13	0.63	0.31	0.87	0.37	0.69	0.37	0.58	0.42	0.45	8
2	0.57	0.68	0.56	0.43	0.32	0.44	0.54	0.61	0.53	0.56	4
3	0.00	0.56	0.00	1.00	0.44	1.00	0.33	0.53	0.33	0.40	9
4	0.47	0.80	0.75	0.53	0.20	0.25	0.48	0.72	0.67	0.62	3
5	0.23	0.90	0.19	0.77	0.10	0.81	0.39	0.84	0.38	0.54	6
6	1.00	0.54	1.00	0.00	0.46	0.00	1.00	0.52	1.00	0.84	1
7	0.60	1.00	0.34	0.40	0.00	0.66	0.55	1.00	0.43	0.66	2
8	0.82	0.17	0.44	0.18	0.83	0.56	0.73	0.38	0.47	0.53	5
9	0.72	0.00	0.22	0.28	1.00	0.78	0.64	0.33	0.39	0.46	7

The response table for GRG for each factor-level combination is shown in table 4 and the graph of the same is shown in figure 5.2. From the figure, it can be concluded that experimental run 6 represents the optimal treatment, providing the highest radish yield on the selected agricultural land. The optimal combination includes trusted seeds with drip irrigation and DAP manure.

Table 4: GRG for factor-level combination

Level	GRG		
	A (Seed)	B (Irrigation)	C (Manure)
1	0.471	**0.580**	**0.607**
2	**0.667**	0.541	0.546
3	0.548	0.565	0.533

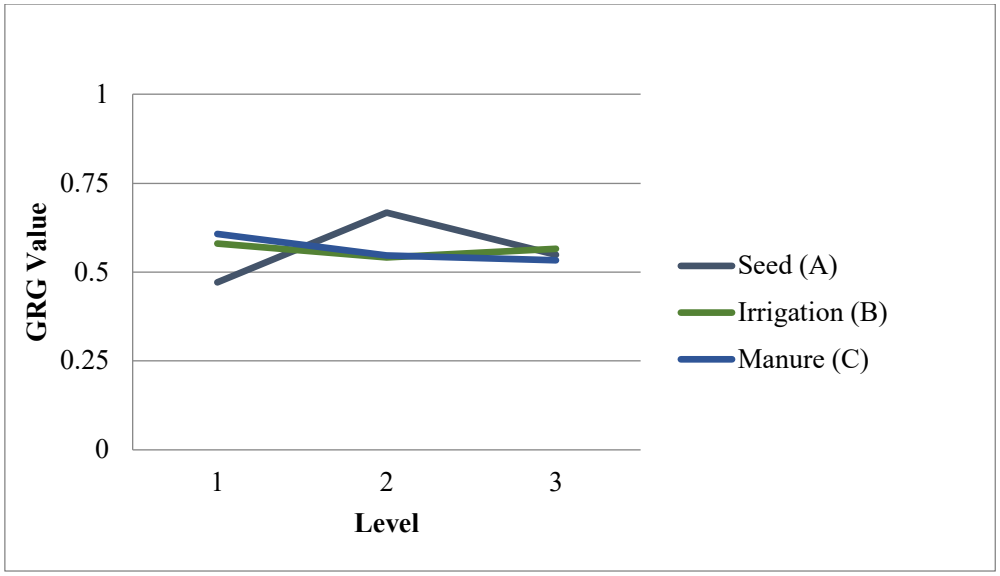


Fig 4: Grey Relational Grades (GRG) by Factor levels

Confirmatory experiments with the optimized levels of the control factors were carried out under similar environmental conditions, maintaining consistency in the plot preparation. This treatment was applied to 4 plots and the data of the three responses was collected. The average radish length of 14.82 cm, quantity of 36 radishes, and weight of 2.31 kg was obtained during the confirmatory experiments which were closely aligning with the experimental findings. This validation strengthens the recommendation that the combination of Trusted seeds, Drip irrigation, and DAP manure represents the optimal treatment for maximizing radish yield under the given agricultural conditions.

Conclusions:

This study systematically evaluated the effects of seed variety, irrigation method, and manure composition on radish growth using an L9 Taguchi experimental design coupled with Grey Relational Analysis (GRA) for multi-response optimization. The results indicate that seed variety is the most critical factor influencing radish performance, followed by manure composition and irrigation method. Varietal selection significantly affected key growth parameters, including root length, yield, and overall quality, underscoring the importance of genetic factors in optimizing crop productivity. Through Grey Relational Analysis, the multi-response optimization identified the combination of Seed variety A2, Irrigation method B1, and Manure composition C1 as the most effective for maximizing growth and yield. This demonstrates that a coordinated approach—integrating appropriate seed selection, organic manure application, and optimal irrigation—can enhance radish performance while promoting sustainable and resource-efficient cultivation practices. Overall, the study provides a robust, quantitative framework for optimizing radish cultivation, minimizing trial-and-error experimentation, and supporting evidence-based agronomic decisions. The findings offer practical guidance for farmers and agricultural practitioners seeking to improve radish yield and quality efficiently. Future research could extend these insights by exploring additional environmental factors, soil types, and management practices to further enhance crop performance across diverse agro-climatic conditions.

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